

Capture The Flag

Mission: Learn how Angular applications are built using reusable components.

Objective: Transform your first Angular application into a modular application using reusable components while following a basic Git workflow.

Flag 0 – Return to Base

Mission

Open the Angular project you created during the previous activity.

Tasks

- Open **Visual Studio Code**.
- Open your Angular project folder.
- Open the integrated terminal.
- Verify that the project still runs.
 - ng serve

Open:

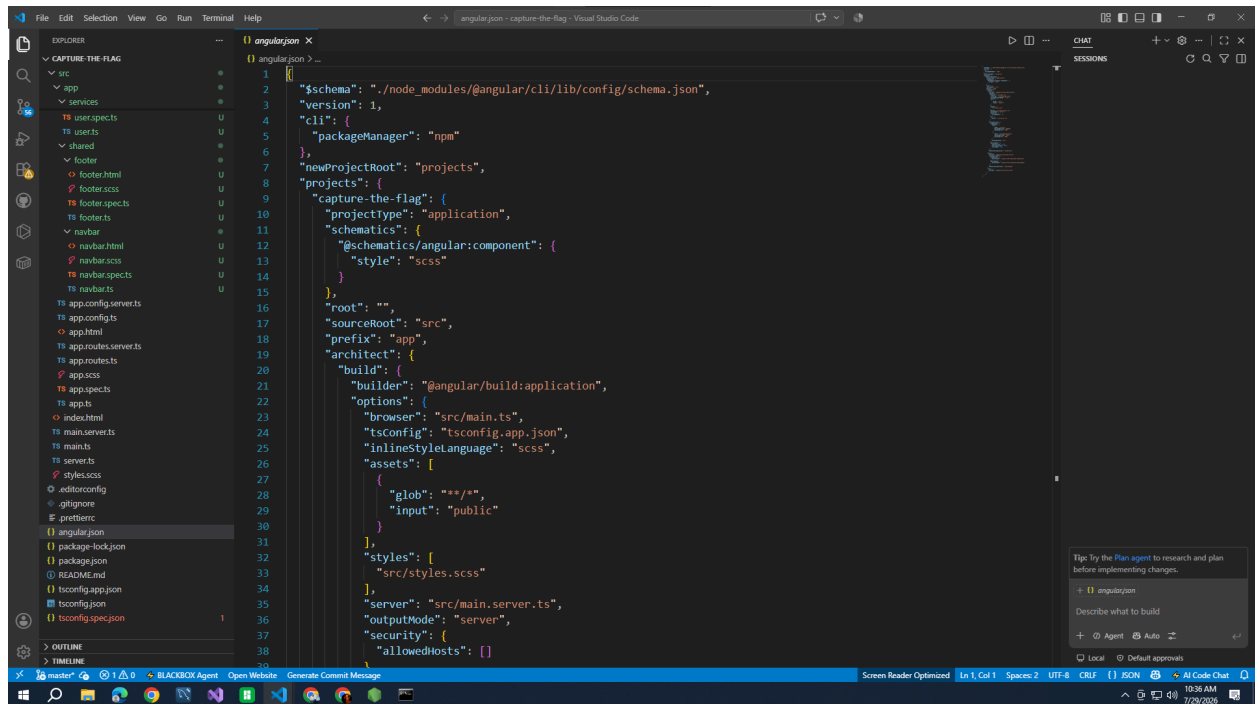
`http://localhost:4200`

If your Angular welcome page appears, proceed.



Evidence

- VS Code with project opened
- Terminal showing `ng serve`
- Browser displaying the Angular application



```
Server bundles
Initial chunk files | Names | Raw size
main.server.mjs | main.server | 49.86 kB |
server.mjs | server | 1.84 kB |
polyfills.server.mjs | polyfills.server | 268 bytes |

Application bundle generation complete. [2.237 seconds] - 2026-07-29T02:45:22.765Z

Watch mode enabled. Watching for file changes...
NOTE: Raw file sizes do not reflect development server per-request transformations.
→ Local: http://localhost:4200/
→ press h + enter to show help
```



Hello, Syvy Sarvida, BSIT-IV

Congratulations! Your app is running. 🎉

Profile Avatar



Flag 1 – Mission Branch

Mission

Every new feature must have its own Git branch.

Create a new branch named:

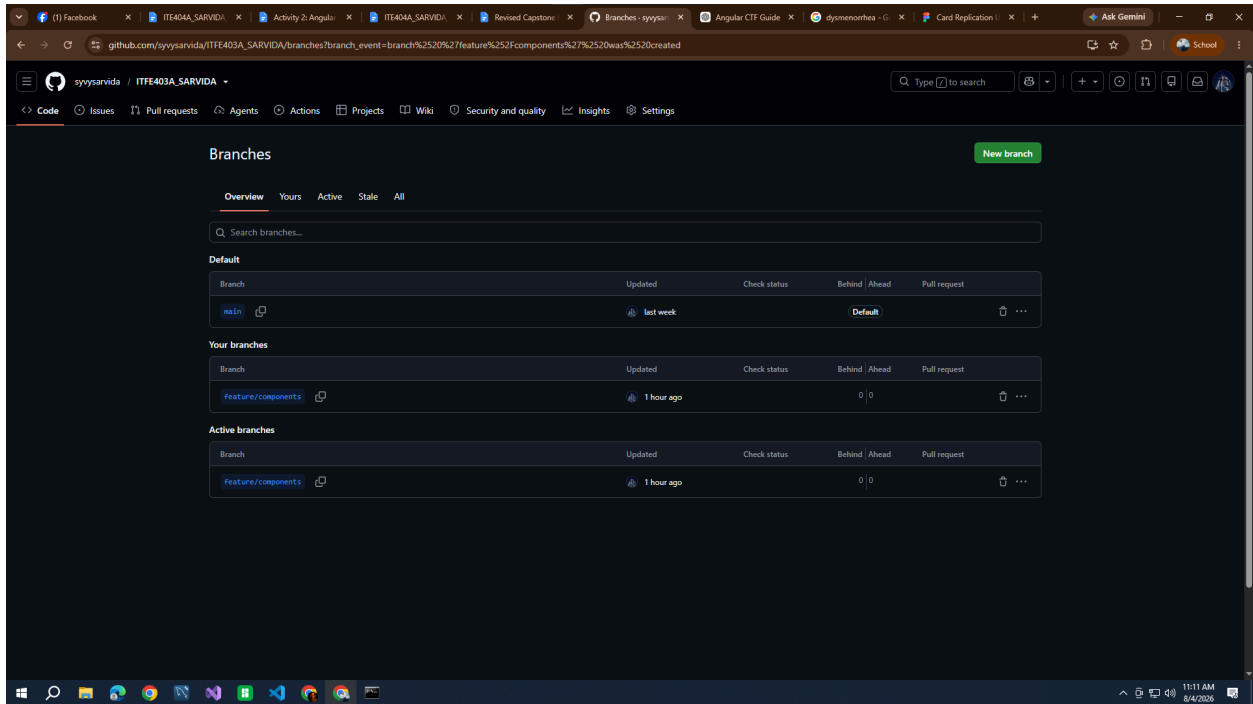
feature/components

Commands:



Evidence

- Terminal showing the successful branch creation
- Output of `git branch`



Flag 2 – Explore the Base

Mission

Investigate your Angular project before making changes.

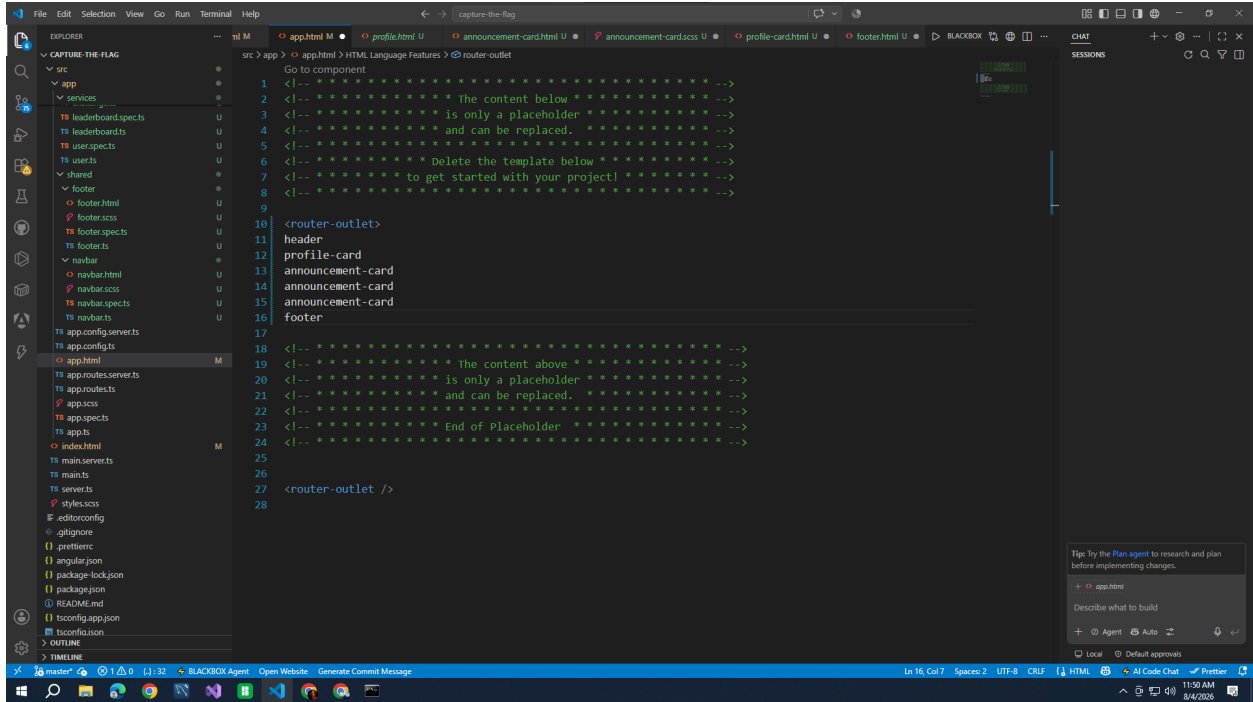
Locate the following:

- app.html
- app.ts
- app.css
- app.spec.ts

Understand that these files belong to the **App Component**, the root component of the application.

Evidence

- Explorer showing the files
- **app.html** opened



Flag 3 – Clear the Battlefield

Mission

Remove the default Angular welcome page.

Delete everything inside:

app.component.html

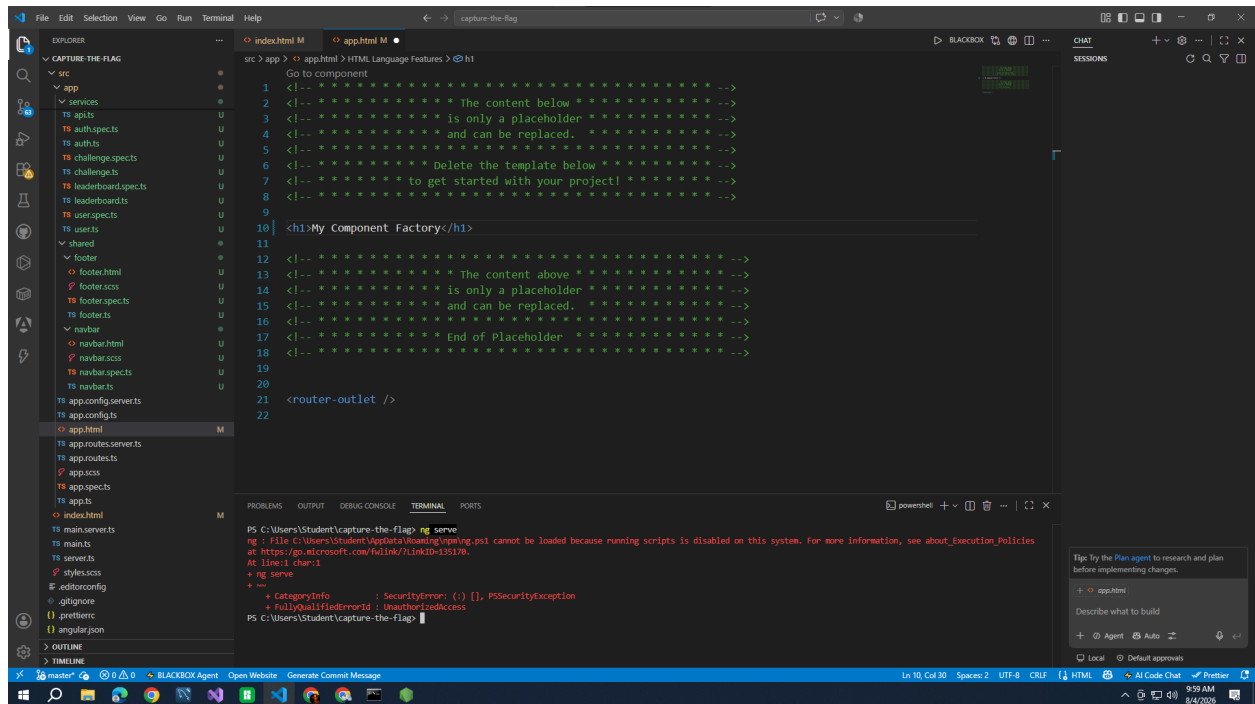
Replace it with:

<h1>My Component Factory</h1>

Verify the browser updates.

Evidence

- Browser showing only the heading
- HTML file



Flag 4 – Build the Headquarters

Mission

Create a reusable Header component.

Generate:

```
ng generate component components/header
```

Angular should automatically generate:

header.ts

header.html

header.css

header.spec.ts

Evidence

- Terminal
- Explorer showing generated files

```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Student>cd C:\Users\Student\capture-the-flag

C:\Users\Student\capture-the-flag>ng generate component components/header
CREATE src/app/components/header/header.spec.ts (554 bytes)
CREATE src/app/components/header/header.ts (198 bytes)
CREATE src/app/components/header/header.scss (0 bytes)
CREATE src/app/components/header/header.html (22 bytes)

C:\Users\Student\capture-the-flag>
```

Flag 5 – Raise the Banner

Mission

Inside the Header component, create a simple header.

Example:

```
<header>
  <h2>Don Bosco Student Portal</h2>
</header>
```

Use the component inside:

```
app.html
<app-header></app-header>
```

Verify it appears in the browser.



Evidence

- Browser
- HTML code

My Component Factory

Don Bosco Student Portal

© 2026 Don Bosco Technical College

Flag 6 – Secure the Exit

Mission

Create a Footer component.

Use the same command previously, but now for the footer.

Display:

```
<footer>
```

```
  © 2026 Don Bosco Technical College
```

```
</footer>
```

Insert it into the App Component.



Evidence

- Browser
- Generated files


```
C:\Users\Student\capture-the-flag>ng g c components/footer
CREATE src/app/components/footer/footer.spec.ts (554 bytes)
CREATE src/app/components/footer/footer.ts (198 bytes)
CREATE src/app/components/footer/footer.scss (0 bytes)
CREATE src/app/components/footer/footer.html (22 bytes)
```

Flag 7 – Create the Profile Card

Mission

Generate a reusable Profile Card.

Use the command to create a new component called profile-card.

Create a card containing:

- Profile Picture
- Name
- Course
- Year Level

(Design is optional)

Insert it into:

app.html

Evidence

- Browser
- Profile Card component

```
C:\Users\Student\capture-the-flag>ng g c components/profile-card
CREATE src/app/components/profile-card/profile-card.spec.ts (590 bytes)
CREATE src/app/components/profile-card/profile-card.ts (221 bytes)
CREATE src/app/components/profile-card/profile-card.scss (0 bytes)
CREATE src/app/components/profile-card/profile-card.html (28 bytes)
```

```
C:\Users\Student\capture-the-flag>ng generate component shared/profile-card
CREATE src/app/shared/profile-card/profile-card.spec.ts (590 bytes)
CREATE src/app/shared/profile-card/profile-card.ts (221 bytes)
CREATE src/app/shared/profile-card/profile-card.scss (0 bytes)
CREATE src/app/shared/profile-card/profile-card.html (28 bytes)
```

```

1 <p>profile-card works!</p>
2 <div class="profile-card">
3   
4
5   <h3>Syvy S. Sarvida</h3>
6
7   <p>Course: BSIT</p>
8
9   <p>Year Level: 4th Year</p>
10 </div>
11

```

Flag 8 – Produce Announcement Cards

Mission

Generate one reusable Announcement Card.

Use the same command, creating the component announcement-card.

Design a card containing:

- Announcement Title
- Description
- Date

Do **NOT** create three separate components.

Instead, reuse the same component.

Example:

```
<app-announcement-card></app-announcement-card>
```

```
<app-announcement-card></app-announcement-card>
```

```
<app-announcement-card></app-announcement-card>
```



Evidence

- Browser
- App component showing three announcement cards

```

C:\Users\Student\capture-the-flag>ng g c components/announcement-card
CREATE src/app/components/announcement-card/announcement-card.spec.ts (625 bytes)
CREATE src/app/components/announcement-card/announcement-card.ts (241 bytes)
CREATE src/app/components/announcement-card/announcement-card.scss (0 bytes)
CREATE src/app/components/announcement-card/announcement-card.html (33 bytes)

```

```
C:\Users\Student\capture-the-flag>ng generate component shared/announcement-card
CREATE src/app/shared/announcement-card/announcement-card.spec.ts (625 bytes)
CREATE src/app/shared/announcement-card/announcement-card.ts (241 bytes)
CREATE src/app/shared/announcement-card/announcement-card.scss (0 bytes)
CREATE src/app/shared/announcement-card/announcement-card.html (33 bytes)
```

Flag 9 – Assemble the Dashboard

Your `app.html` should now act only as the page assembler.

Assemble your page to have this structure:

header

profile-card

announcement-card

announcement-card

announcement-card

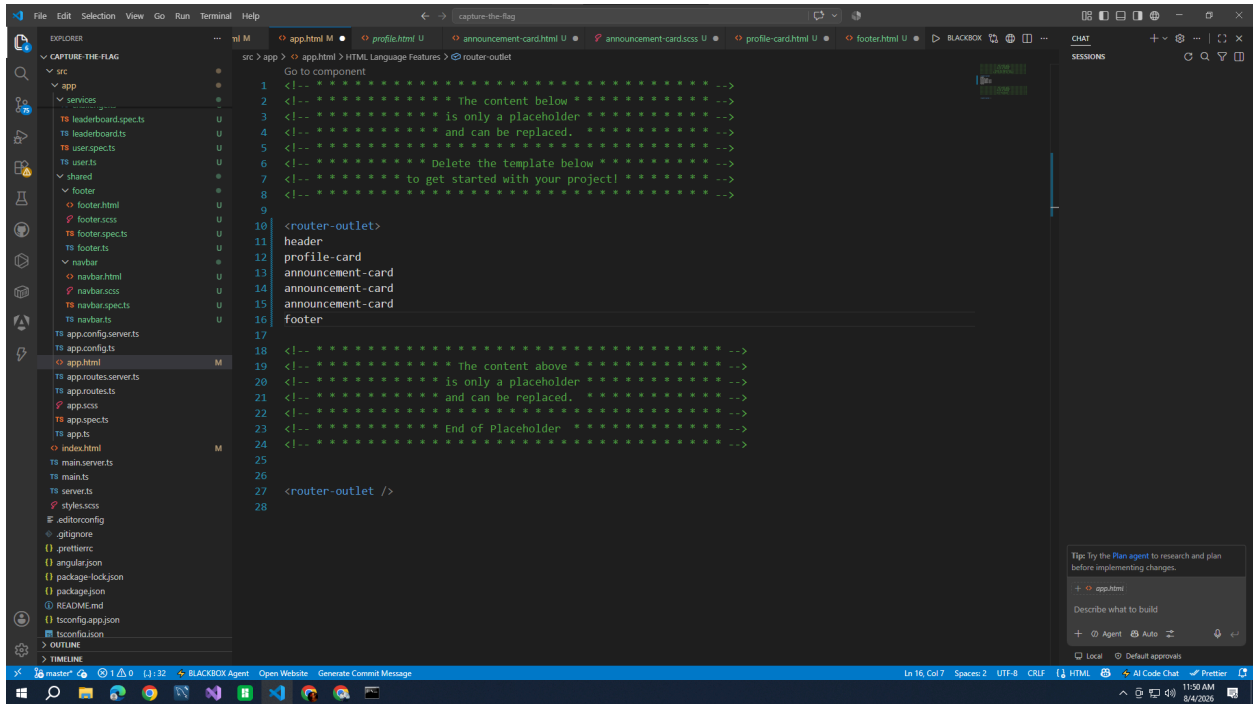
footer

Notice that almost no actual page content exists here. Only reusable components.



Evidence

- Browser
- `App.html`



Flag 10 – Client Changed Their Mind

Mission

The client sends an urgent request:

"Every announcement card should now have:

- Rounded corners
- A blue border
- Padding
- A subtle shadow"

You must update **only one file**.

Apply your styles there and verify that **all announcement cards** update automatically.



Evidence

- Browser before and after the CSS change
- `Announcement-card.css` code

```

1  .announcement-card {
2      border: 2px solid blue;
3      border-radius: 12px;
4      padding: 16px;
5      margin: 10px 0;
6      box-shadow: 0 2px 8px rgba(0,0,0,0.2);
7  }
8

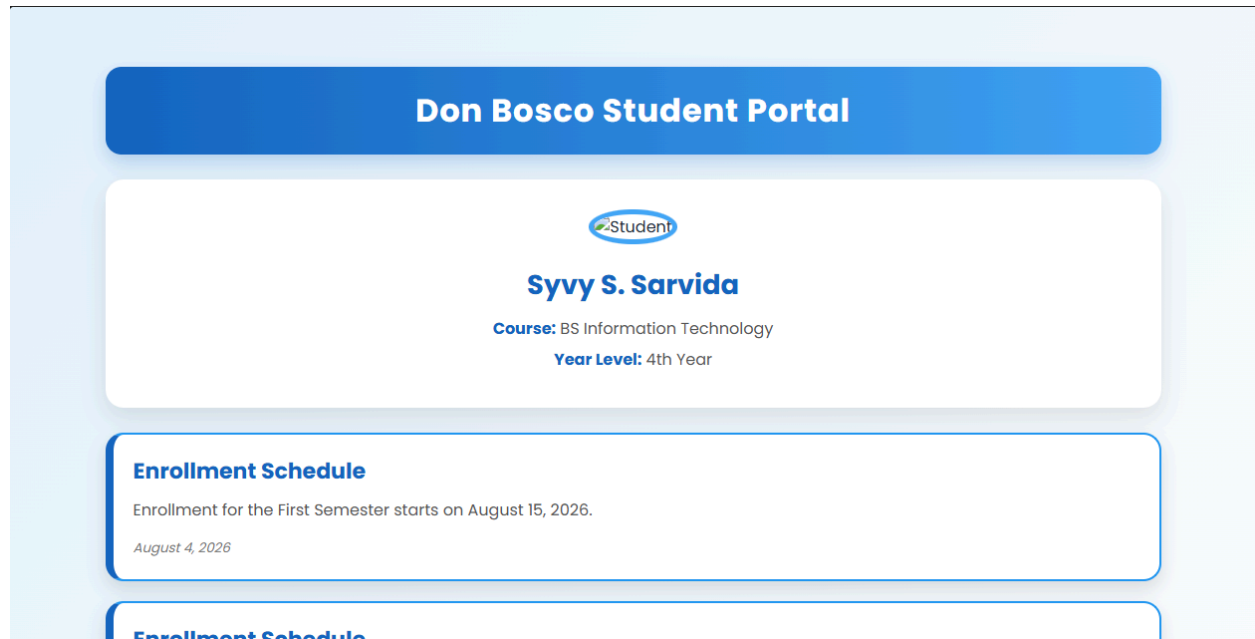
```

Flag 11 – Pushing Changes

Commit your completed work, then push to your repository.

Evidence

- Terminal commands for committing + pushing
- Screenshot of the Github Repository reflecting the pushed changes



```

# Please enter the commit message for your changes. Lines starting
# with '#' will be ignored, and an empty message aborts the commit.
#
# On branch master
# Changes to be committed:
#   new file:   public/4d865ea47a8675d682ff35ad904a0af6.png
#   new file:   public/unnamed.jpg
#   modified:   src/app/app.html
#   modified:   src/app/app.ts
#   new file:   src/app/components/announcement-card/announcement-card.component.ts
#   new file:   src/app/components/announcement-card/announcement-card.html
#   new file:   src/app/components/announcement-card/announcement-card.scss
#   new file:   src/app/components/announcement-card/announcement-card.spec.ts
#   new file:   src/app/components/footer/footer.html
#   new file:   src/app/components/footer/footer.scss
#   new file:   src/app/components/footer/footer.spec.ts
#   new file:   src/app/components/footer/footer.ts
#   new file:   src/app/components/header/header.html
#   new file:   src/app/components/header/header.scss
#   new file:   src/app/components/header/header.spec.ts
#   new file:   src/app/components/header/header.ts
#   new file:   src/app/components/profile-card/profile-card.component.ts
#   new file:   src/app/components/profile-card/profile-card.html
#   new file:   src/app/components/profile-card/profile-card.scss
#   new file:   src/app/components/profile-card/profile-card.spec.ts

```

